

(1) t vs. $\delta z(t)$ with steady-state identification (accompanied by a description of steady-state criterion used), and helix snapshots (≥ 5) with labeled axes/units/time:

Figure 1 shows the axial tip displacement $\delta z(t)$ under a constant tensile load $F=F_{\text{char}}$. The transient response is oscillatory due to inertia and structural flexibility, with clear exponential decay. To identify steady state, I applied the criterion:

$$\frac{|\delta z(t) - \delta z(t-1)|}{|\delta z(t)|} < 1\%$$

The displacement stabilizes around:

$$\delta z^* = -0.0015523549747709614 \text{ (m)}$$

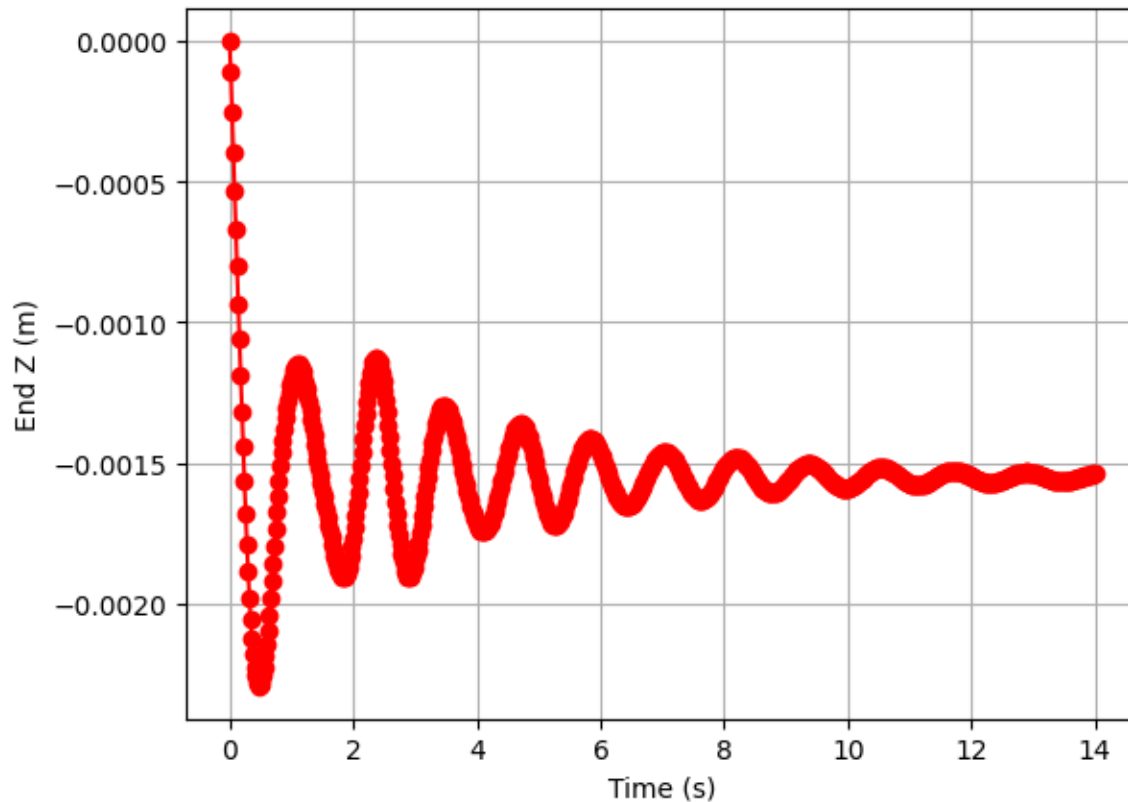
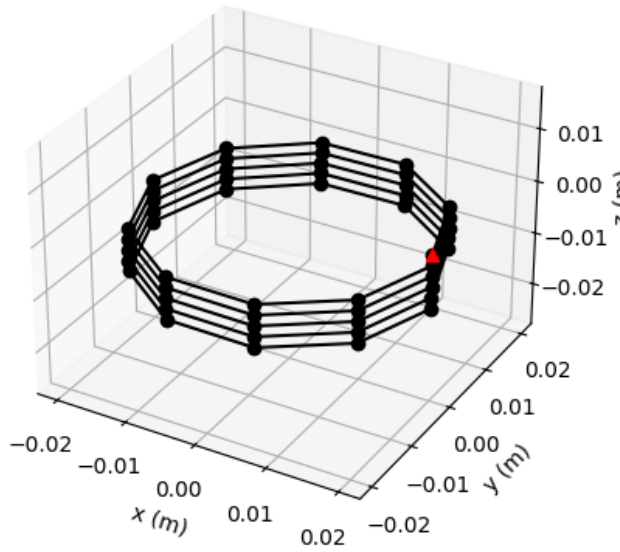


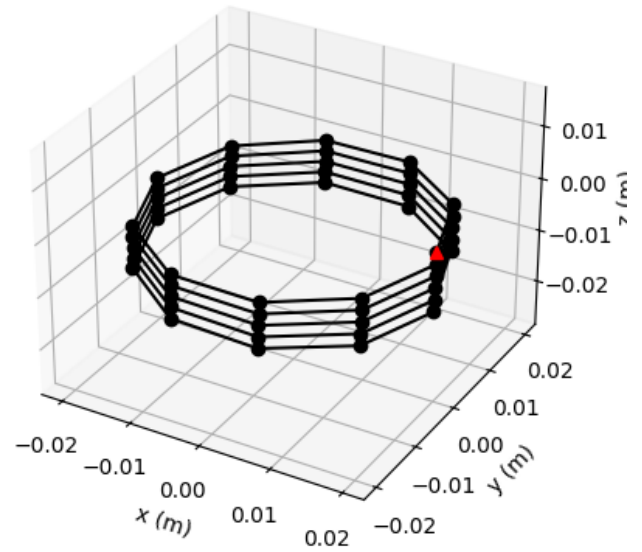
Figure 1. Time history of axial displacement $\delta z(t)$ under constant load $F=F_{\text{char}}$. Steady state is identified when changes over a 1 s window fall below 1%.

Figures 2(a–e) show five snapshots of the helix configuration at increasing times ($t = 0, 3.70, 7.45, 11.20, 14.95$ s). All axes are labeled and scaled equally. As the simulation progresses, the helix stretches downward and then settles into a steady configuration while maintaining its overall geometry (because the displacements are small an obvious change in the spring isn't seen here).

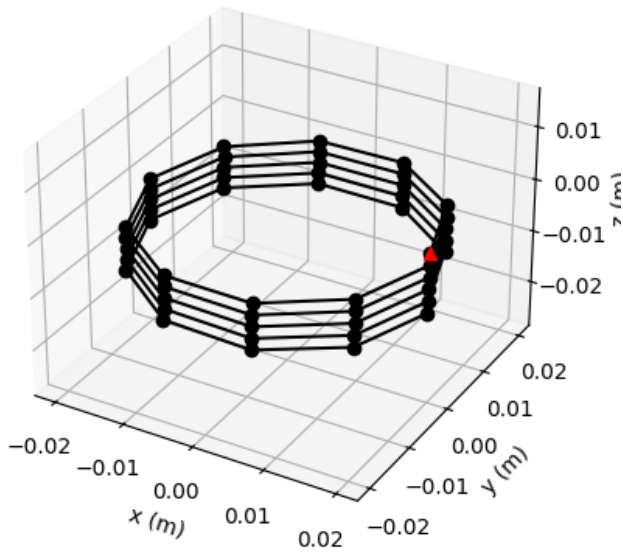
a) $t = 0.00$ s



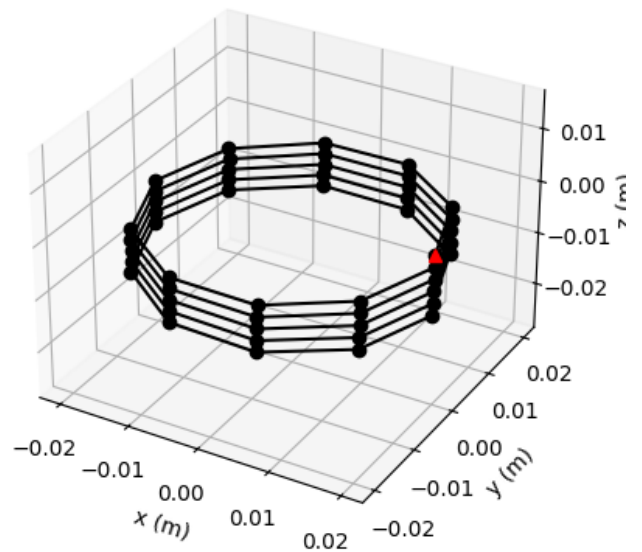
b) $t = 3.48$ s



c) $t = 6.98$ s



d) $t = 10.48$ s



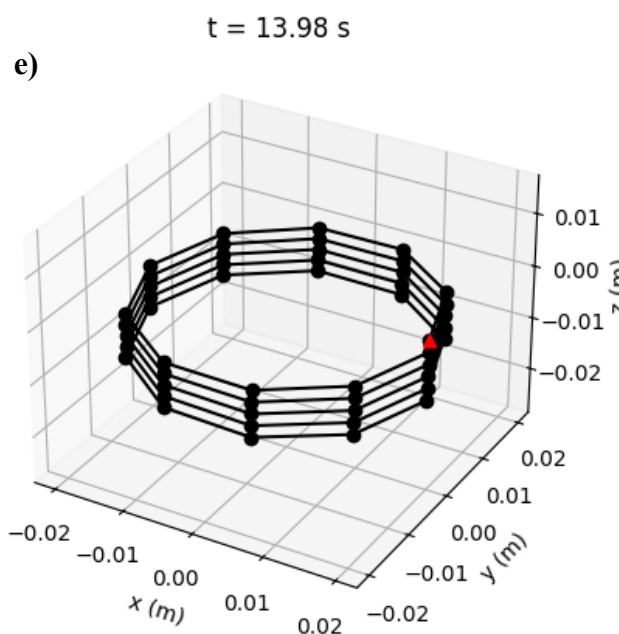


Figure 2. Five snapshots of the helix at (a) $t = 0.0 \text{ s}$, (b) $t = 3.48 \text{ s}$, (c) $t = 6.98 \text{ s}$, (d) $t = 10.48 \text{ s}$, (e) $t = 13.98 \text{ s}$

(2) F vs. δz with zero-intercept best-fit line:

The force sweep results you obtained:

$F \text{ (N)}$	$\delta z^* \text{ (m)}$
$1.989\text{e-}07$	$1.5598\text{e-}05$
$5.336\text{e-}07 \text{ N}$	$4.1841\text{e-}05$
$1.431\text{e-}06$	$1.1222\text{e-}04$
$3.840\text{e-}06$	$3.0091\text{e-}04$
$1.030\text{e-}05$	$8.0603\text{e-}04$
$2.764\text{e-}05$	$2.1522\text{e-}03$
$7.414\text{e-}05$	$5.6766\text{e-}03$
$1.989\text{e-}04$	$1.4248\text{e-}02$

Table 1. Numerical values of δz^* for each F with a constant F_{char}

The best-fit line through the origin, using the relation:

$$F = k * \delta z$$

gives:

$$k = 0.0127605356877123 \text{ N/m}$$

Figure 3 shows the simulation points and the fitted line. The linear region is extremely well-behaved, confirming the simulation captures the expected small-displacement spring behavior.

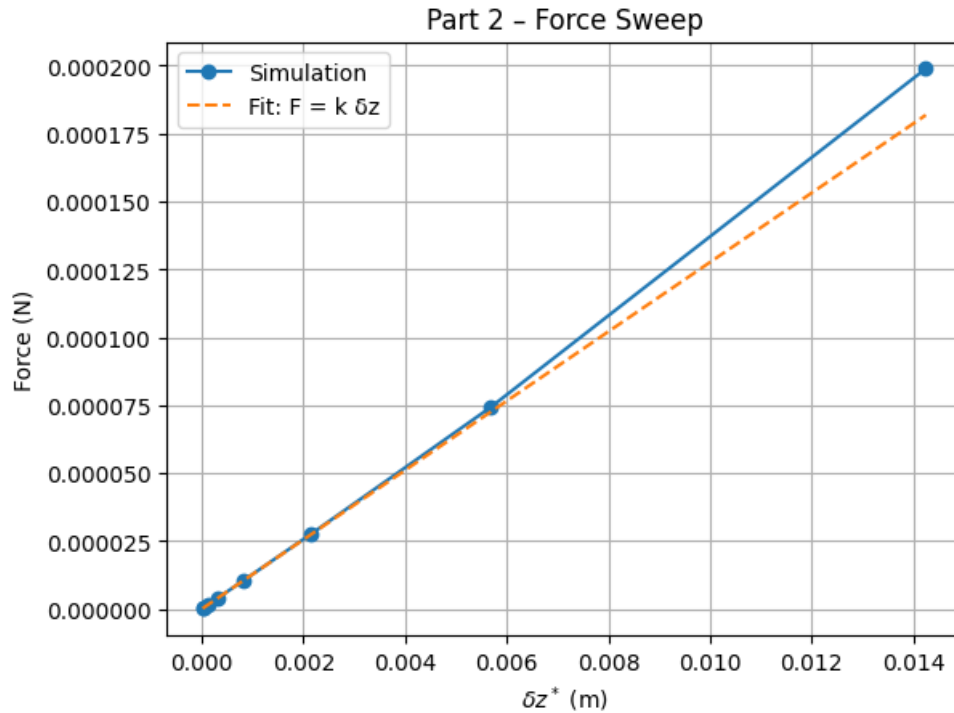
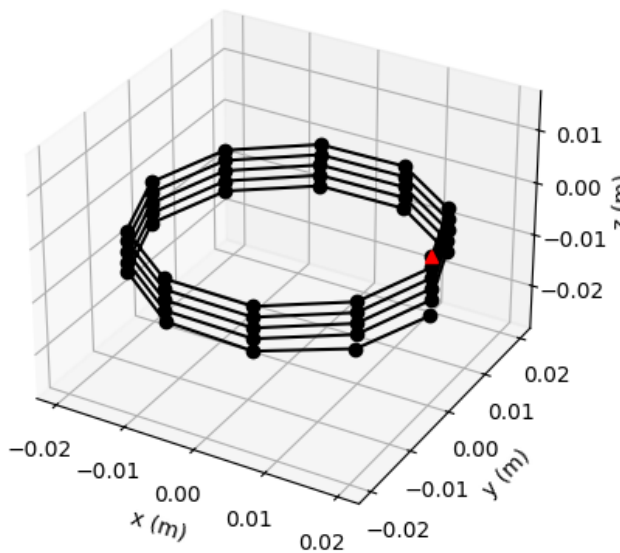


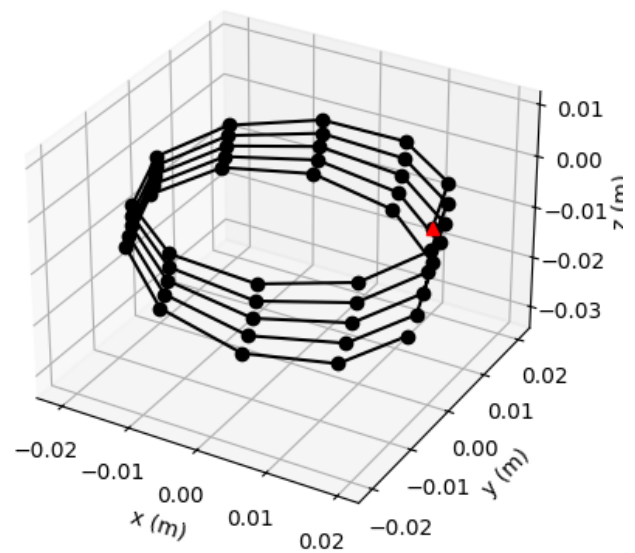
Figure 3. Force–displacement curve from Part 2 showing simulation data and the zero-intercept linear fit $F=k \delta z^*$.

Also, a sample of the spring displacement is included here for the max force of $10F_{\text{char}}$:

a) $F = 1.989\text{e-}04 \text{ N}$, $t = 0.00 \text{ s}$



b) $F = 1.989\text{e-}04 \text{ N}$, $t = 3.48 \text{ s}$



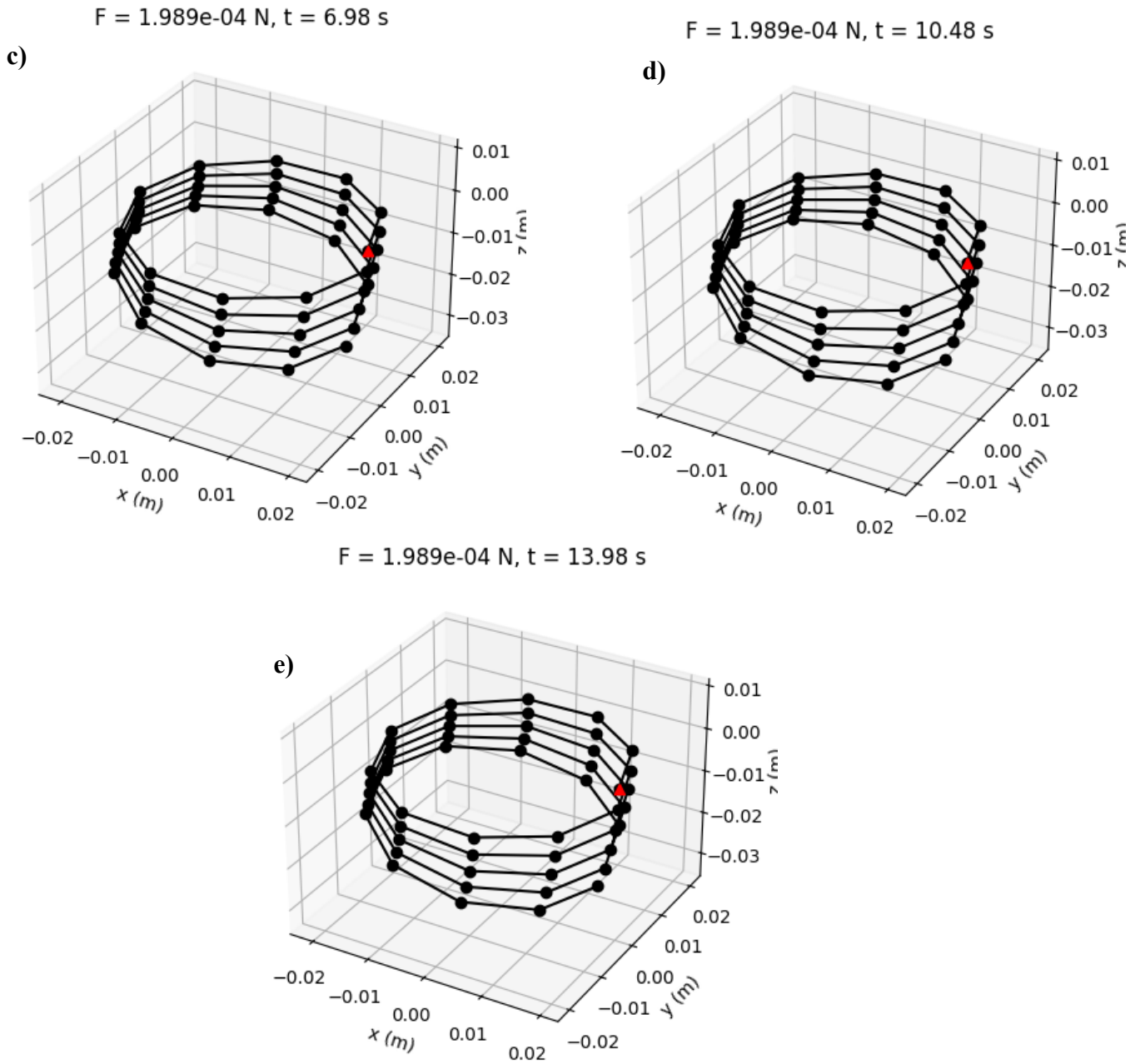


Figure 4. Five snapshots of the helix at (a) $t = 0.0 \text{ s}$, (b) $t = 3.48 \text{ s}$, (c) $t = 6.98 \text{ s}$, (d) $t = 10.48 \text{ s}$, (e) $t = 13.98 \text{ s}$ with $F = 10F_{\text{Char}}$

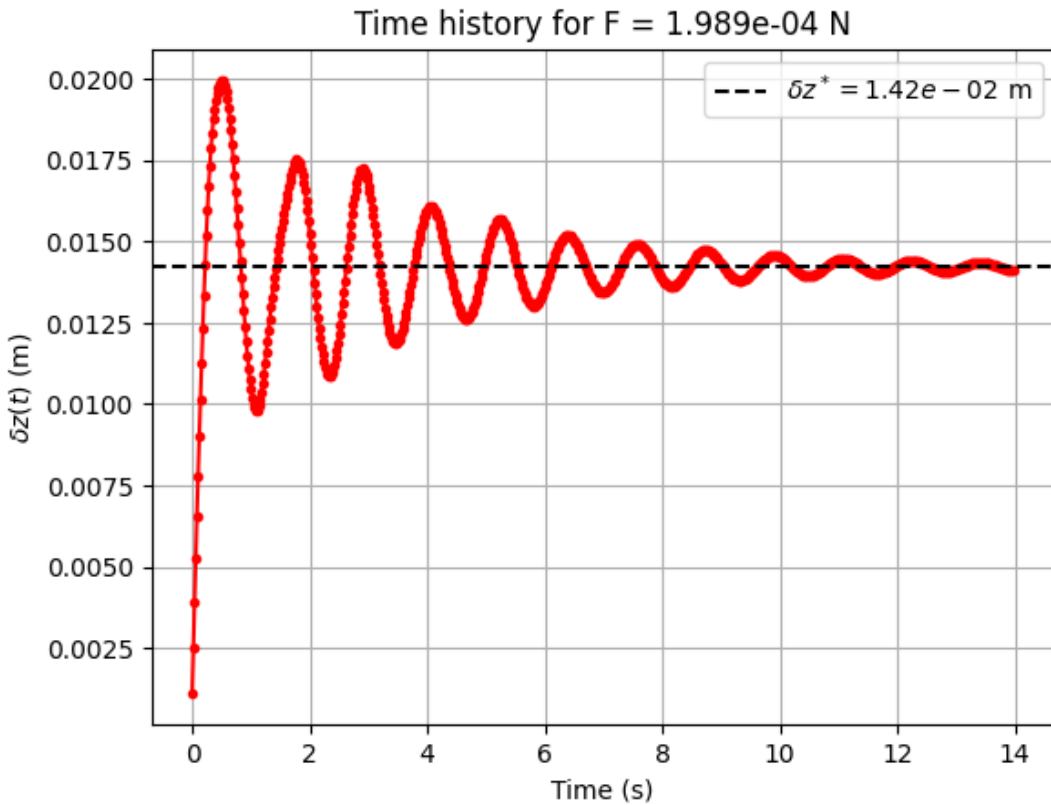


Figure 5. Time history of axial displacement $\delta z(t)$ under constant load $F=10F_{\text{char}}$. (For a better look, the change in length has been plotted with a positive sign)

(3) k vs. $Gd^4/(8ND^3)$ with the slope-1 reference:

By varying the coil diameter D from 0.01 m to 0.05 m. The corresponding DER-computed stiffnesses were found:

D (m)	K (N/m)
0.010	1.074e+00
0.0144	3.998e-01
0.0189	1.947e-01
0.0233	1.099e-01
0.0278	6.860e-02
0.0322	4.616e-02
0.0367	3.293e-02

0.0411	2.458e-02
0.0456	1.902e-02
0.0500	1.512e-02

Table 2. Numerical values of **k** for each **D** with a varying **F_{char}**.

The textbook close-coiled approximation predicts:

$$K_{text} = \frac{Gd^4}{8ND^3}$$

Figure 6, the plot of DER stiffness $k(D)$ versus the textbook quantity $Gd^4/(8ND^3)$ shows:

- a strong positive correlation,
- a generally linear trend,
- and the correct qualitative dependence:

larger helix diameter $\Rightarrow$ more flexible spring $\Rightarrow$ k decreases

The general trend is correct but the DER data fall below the slope-1 line at larger D . This is expected because:

1. The textbook formula assumes
 - small deformation
 - thin wire
 - close-coiled geometry
 - pure torsion in the wire.
2. The helix has finite pitch, bending contributions, non-uniform strain, and 3D elastic rod behavior that is not captured in the classical closed-form approximation.

Thus, qualitative agreement is correct, and quantitative mismatch is expected, exactly as the homework file states.

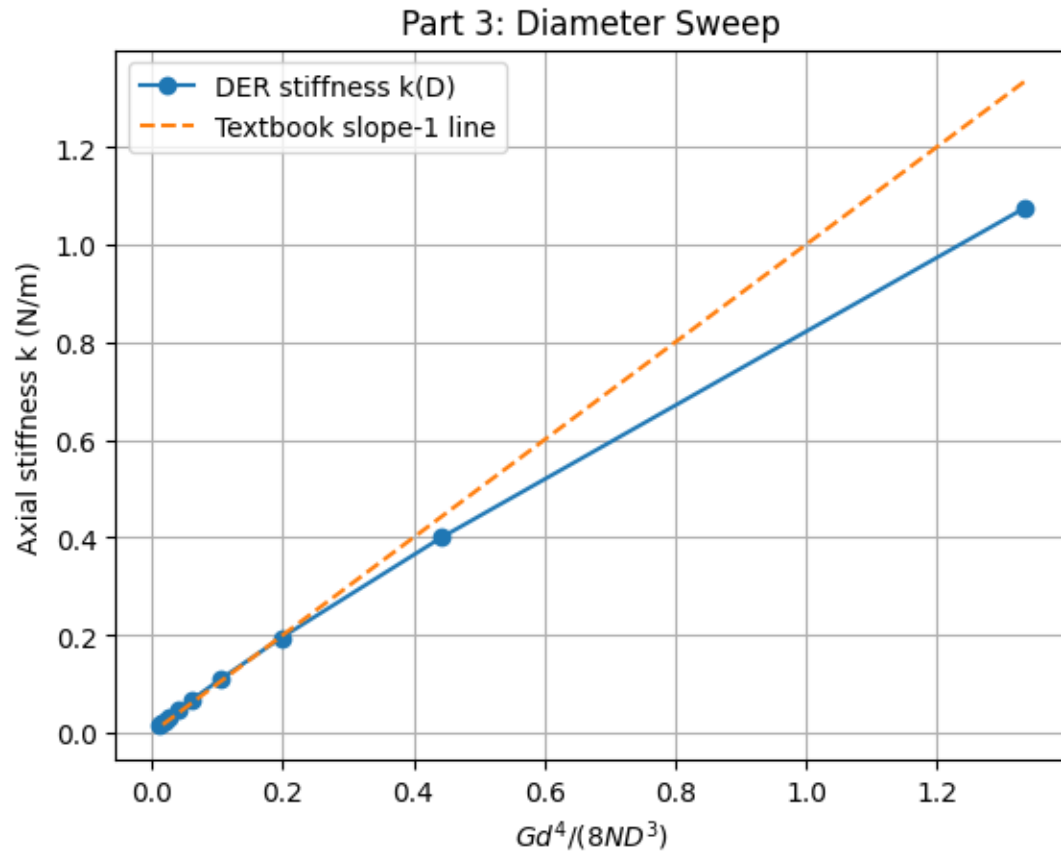


Figure 6. DER-computed stiffness $k(D)$ versus the textbook torsional spring quantity $Gd^4/(8ND^3)$. The dashed line is the classical prediction $K_{\text{text}}=x$ (slope 1).